

A

B

C

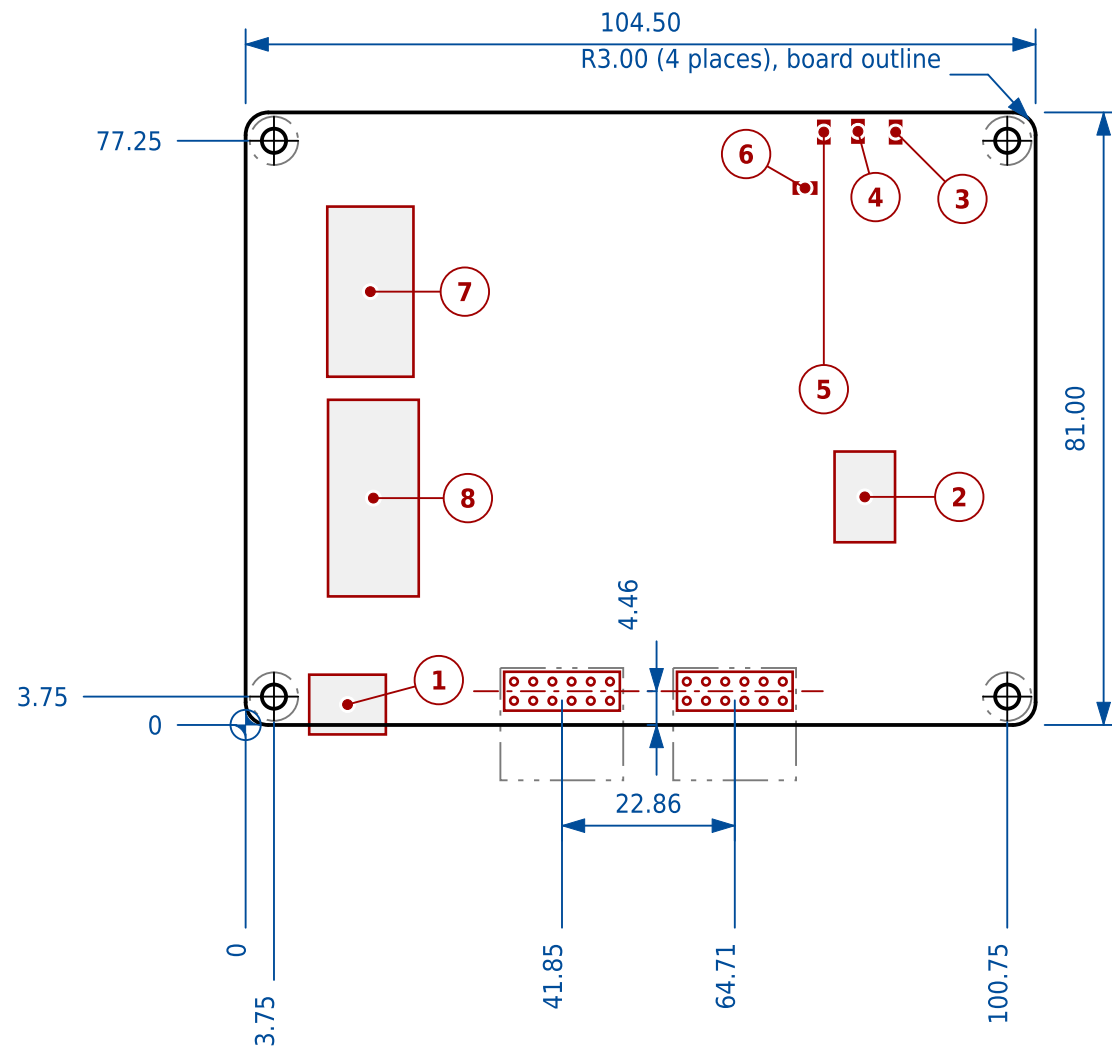
D

A

B

C

D



### NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
- Geometry is design nominal, read from the KiCad board file.
- Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
- The title block reads "TinyTapeout 1,2,3 Demo Board", but no PCB served TT01: that was a bare-die trial run.
- This sheet covers revisions 2.2.5 and 2.2.6: the same board mechanically, compared feature by feature before merging. Only the electrical design and the shuttle differ.
- The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0, is the same on every revision; their distance from the lower edge and their position along it are not.
- Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.
- Board thickness is 1.6 mm nominal. The KiCad file's 1.60000 mm is its stackup sum, not a specified finished thickness.

### NOTES (continued)

- Assembled envelope, connector overhang included, is 104.50 x 82.25 mm against a 104.50 x 81.00 mm board outline. The FEATURE SCHEDULE gives the extents.
- GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
- Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

### SOURCES

- KiCad board file: [https://github.com/TinyTapeout/tt123-demo-pcb/mpw-mb1.kicad\\_pcb](https://github.com/TinyTapeout/tt123-demo-pcb/mpw-mb1.kicad_pcb) @ 303509a - title block: TinyTapeout 1,2,3 Demo Board rev 2.2.5, dated 2023-11-13
- KiCad board file: [https://github.com/TinyTapeout/tt123-demo-pcb/mpw-mb1.kicad\\_pcb](https://github.com/TinyTapeout/tt123-demo-pcb/mpw-mb1.kicad_pcb) @ 3d721a6 - title block: TinyTapeout 1,2,3 Demo Board rev 2.2.6, dated 2024-02-14
- Shuttle mapping: [https://docs.google.com/spreadsheets/d/1xD\\_uemuQcpUnXRY4twOrFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370](https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370) - used by: TT02, TT03

### HOLE SCHEDULE

| ID  | X mm   | Y mm  | DIA mm | KEEPOUT mm |
|-----|--------|-------|--------|------------|
| MT1 | 3.75   | 3.75  | 3.20   | 6.40       |
| MT2 | 100.75 | 3.75  | 3.20   | 6.40       |
| MT3 | 3.75   | 77.25 | 3.20   | 6.40       |
| MT4 | 100.75 | 77.25 | 3.20   | 6.40       |

### PMOD HOST HEADERS

| PORT   | REF | EDGE   | CX mm  | CY mm | PIN1 X mm | PIN1 Y mm |
|--------|-----|--------|--------|-------|-----------|-----------|
| PMOD A | J3  | bottom | 41.850 | 4.465 | 48.200    | 5.735     |
| PMOD B | J9  | bottom | 64.710 | 4.465 | 71.060    | 5.735     |

### FEATURE SCHEDULE

| #  | FEATURE                                        | X EXTENT mm    | Y EXTENT mm    |
|----|------------------------------------------------|----------------|----------------|
| 1  | USB-C power / control                          | 8.41 to 18.55  | -1.25 to 6.65  |
| 2  | 7-segment display                              | 77.91 to 85.91 | 24.16 to 36.16 |
| 3  | LED D1                                         | 85.25 to 86.72 | 76.92 to 79.88 |
| 4  | LED D2                                         | 80.27 to 81.73 | 77.02 to 79.98 |
| 5  | LED D3                                         | 75.75 to 77.22 | 76.92 to 79.88 |
| 6  | LED D4                                         | 72.52 to 75.48 | 70.27 to 71.73 |
| 7  | 8-way input DIP switch                         | 10.80 to 22.20 | 46.05 to 68.55 |
| 8  | 9-way DIP switch SW2                           | 10.90 to 22.90 | 17.00 to 43.00 |
| 9  | Side Pmod, first position - not on this board  | -              | -              |
| 10 | Side Pmod, second position - not on this board | -              | -              |
| 11 | Side Pmod, third position - not on this board  | -              | -              |

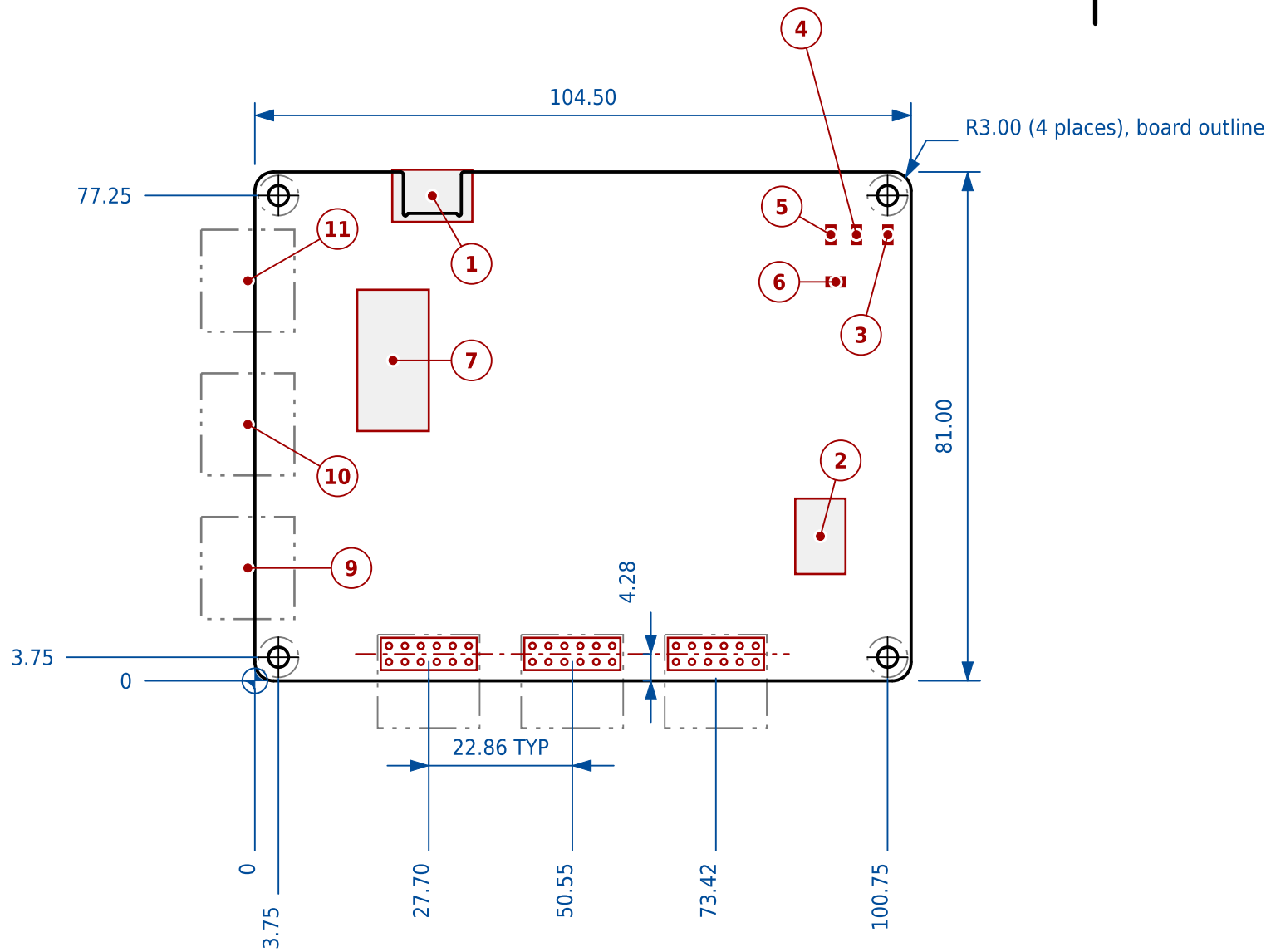
### LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body, hole keep-out
- Dimension, extension and leader

## DB MPW V2.2.5 / DB MPW V2.2.6

mpw-mb1 rev 2.2.5 and 2.2.6

|                  |                   |                                                |       |
|------------------|-------------------|------------------------------------------------|-------|
| DRAWN            | VERSION           | UNITS                                          | SCALE |
| generated        | v0.0-98-g6a42ae3+ | mm                                             | 1:1   |
| SIZE             | SHEET             | DRAWING NO                                     | REV   |
| A3               | 1 OF 1            | TT-DB-01                                       | A     |
| MATERIAL         |                   | GENERAL TOLERANCE                              |       |
| PCB, 1.6 nominal |                   | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |       |



## NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
- Geometry is design nominal, read from the KiCad board file.
- Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
- v1.2.2c is an assembly variant, not a layout change: no X1, a 0R at R51. Its KiCad title block reads rev 1.2.3.
- This sheet covers revisions 1.2.2 and 1.2.3: the same board mechanically, compared feature by feature before merging. Only the electrical design and the shuttle differ.
- The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0, is the same on every revision; their distance from the lower edge and their position along it are not.
- Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.
- Board thickness is 1.6 mm nominal. The KiCad file's 1.56252 mm is its stackup sum, not a specified finished thickness.

## NOTES (continued)

- Assembled envelope, connector overhang included, is 104.50 x 81.35 mm against a 104.50 x 81.00 mm board outline. The FEATURE SCHEDULE gives the extents.
- Upper edge carries a shallow recess for the USB-C shell, contributed by the connector footprint's own edge cuts.
- GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
- Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

## SOURCES

- KiCad board file: [https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad\\_pcb](https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad_pcb) @ cfdd80d7b - title block: TinyTapeout 4+ Demoboard rev 1.2.2, dated 2024-04-26
- KiCad board file: [https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad\\_pcb](https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad_pcb) @ a88cbc08b - title block: TinyTapeout 4+ Demoboard rev 1.2.3, dated 2024-06-11
- Shuttle mapping: [https://docs.google.com/spreadsheets/d/1xD\\_uemuQcpUnXRY4twOrFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370](https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370) - used by: TT04, TT05

## HOLE SCHEDULE

| ID  | X mm   | Y mm  | DIA mm | KEEPOUT mm |
|-----|--------|-------|--------|------------|
| MT1 | 3.75   | 3.75  | 3.20   | 6.40       |
| MT2 | 100.75 | 3.75  | 3.20   | 6.40       |
| MT3 | 3.75   | 77.25 | 3.20   | 6.40       |
| MT4 | 100.75 | 77.25 | 3.20   | 6.40       |

## PMOD HOST HEADERS

| PORT   | REF | EDGE   | CX mm  | CY mm | PIN1 X mm | PIN1 Y mm |
|--------|-----|--------|--------|-------|-----------|-----------|
| INPUT  | J3  | bottom | 27.695 | 4.275 | 34.045    | 5.545     |
| BIDIR  | J5  | bottom | 50.555 | 4.275 | 56.905    | 5.545     |
| OUTPUT | J6  | bottom | 73.415 | 4.275 | 79.765    | 5.545     |

## FEATURE SCHEDULE

| #  | FEATURE                               | X EXTENT mm      | Y EXTENT mm    |
|----|---------------------------------------|------------------|----------------|
| 1  | USB-C power / control                 | 21.88 to 34.62   | 73.05 to 81.35 |
| 2  | 7-segment display                     | 86.05 to 94.05   | 17.00 to 29.00 |
| 3  | LED D1                                | 100.07 to 101.53 | 69.52 to 72.48 |
| 4  | LED D2                                | 95.07 to 96.53   | 69.52 to 72.48 |
| 5  | LED D3                                | 90.97 to 92.43   | 69.52 to 72.48 |
| 6  | LED D4                                | 91.02 to 93.98   | 62.77 to 64.23 |
| 7  | 8-way input DIP switch                | 16.30 to 27.70   | 39.75 to 62.25 |
| 8  | Second DIP switch - not on this board | -                | -              |
| 9  | Side Pmod J12 (not fitted)            | -8.55 to 6.30    | 9.83 to 26.08  |
| 10 | Side Pmod J13 (not fitted)            | -8.55 to 6.30    | 32.69 to 48.94 |
| 11 | Side Pmod J14 (not fitted)            | -8.55 to 6.30    | 55.55 to 71.80 |

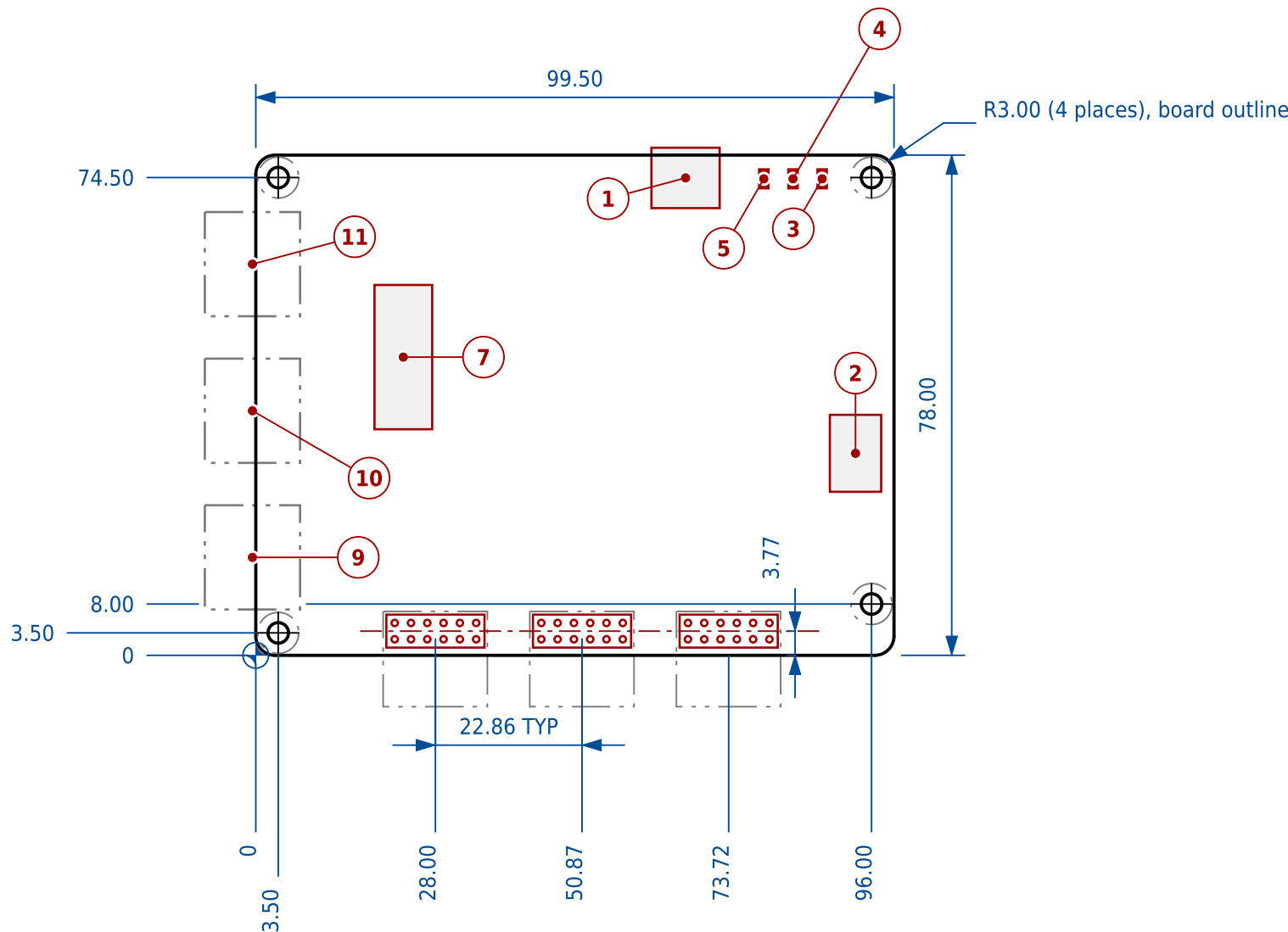
## LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Pmod connector body, hole keep-out
- Dimension, extension and leader

## DB 4+ V1.2.2 / DB 4+ V1.2.2C

tinytapeout-demo rev 1.2.2 and 1.2.3

|                  |                   |                                                |       |
|------------------|-------------------|------------------------------------------------|-------|
| DRAWN            | VERSION           | UNITS                                          | SCALE |
| generated        | v0.0-98-g6a42ae3+ | mm                                             | 1:1   |
| SIZE             | SHEET             | DRAWING NO                                     | REV   |
| A3               | 1 OF 1            | TT-DB-02                                       | A     |
| MATERIAL         |                   | GENERAL TOLERANCE                              |       |
| PCB, 1.6 nominal |                   | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |       |



## NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
- Geometry is design nominal, read from the KiCad board file.
- Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
- This sheet covers revisions 2.0.1 and 2.1.0: the same board mechanically, compared feature by feature before merging. Only the electrical design and the shuttle differ.
- The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0, is the same on every revision; their distance from the lower edge and their position along it are not.
- Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.
- Board thickness is 1.6 mm nominal. The KiCad file's 1.56252 mm is its stackup sum, not a specified finished thickness.

## NOTES (continued)

- Assembled envelope, connector overhang included, is 99.50 x 79.15 mm against a 99.50 x 78.00 mm board outline. The FEATURE SCHEDULE gives the extents.
- GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
- Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

## SOURCES

- KiCad board file: [https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad\\_pcb](https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad_pcb) @ 292760e1f - title block: TinyTapeout 06+ Demoboard rev 2.0.1, dated 2024-09-05
- KiCad board file: [https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad\\_pcb](https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad_pcb) @ a799acb38 - title block: TinyTapeout 06+ Demoboard rev 2.1.0, dated 2024-12-10
- Shuttle mapping: [https://docs.google.com/spreadsheets/d/1xD\\_uemuQcpUnXRY4twOrFzF0urJLVWMdqAnrOJLetM/edit?gid=1250963370](https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrFzF0urJLVWMdqAnrOJLetM/edit?gid=1250963370) - used by: TT06, TT07

## HOLE SCHEDULE

| ID  | X mm  | Y mm  | DIA mm | KEEPOUT mm |
|-----|-------|-------|--------|------------|
| MT1 | 3.50  | 3.50  | 3.20   | 6.40       |
| MT2 | 96.00 | 8.00  | 3.20   | 6.40       |
| MT3 | 3.50  | 74.50 | 3.20   | 6.40       |
| MT4 | 96.00 | 74.50 | 3.20   | 6.40       |

## PMOD HOST HEADERS

| PORT   | REF | EDGE   | CX mm  | CY mm | PIN1 X mm | PIN1 Y mm |
|--------|-----|--------|--------|-------|-----------|-----------|
| INPUT  | J3  | bottom | 28.005 | 3.775 | 34.355    | 5.045     |
| BIDIR  | J5  | bottom | 50.865 | 3.775 | 57.215    | 5.045     |
| OUTPUT | J6  | bottom | 73.725 | 3.775 | 80.075    | 5.045     |

## FEATURE SCHEDULE

| #  | FEATURE                                  | X EXTENT mm    | Y EXTENT mm    |
|----|------------------------------------------|----------------|----------------|
| 1  | USB-C power / control                    | 61.68 to 72.32 | 69.73 to 79.15 |
| 2  | 7-segment display                        | 89.50 to 97.50 | 25.50 to 37.50 |
| 3  | LED D1                                   | 87.57 to 89.03 | 72.82 to 75.78 |
| 4  | LED D2                                   | 83.02 to 84.48 | 72.82 to 75.78 |
| 5  | LED D3                                   | 78.47 to 79.93 | 72.82 to 75.78 |
| 6  | Fourth indicator LED - not on this board | -              | -              |
| 7  | 8-way input DIP switch                   | 18.50 to 27.50 | 35.25 to 57.75 |
| 8  | Second DIP switch - not on this board    | -              | -              |
| 9  | Side Pmod J12 (not fitted)               | -7.95 to 6.90  | 7.15 to 23.40  |
| 10 | Side Pmod J13 (not fitted)               | -7.95 to 6.90  | 30.01 to 46.26 |
| 11 | Side Pmod J14 (not fitted)               | -7.95 to 6.90  | 52.87 to 69.12 |

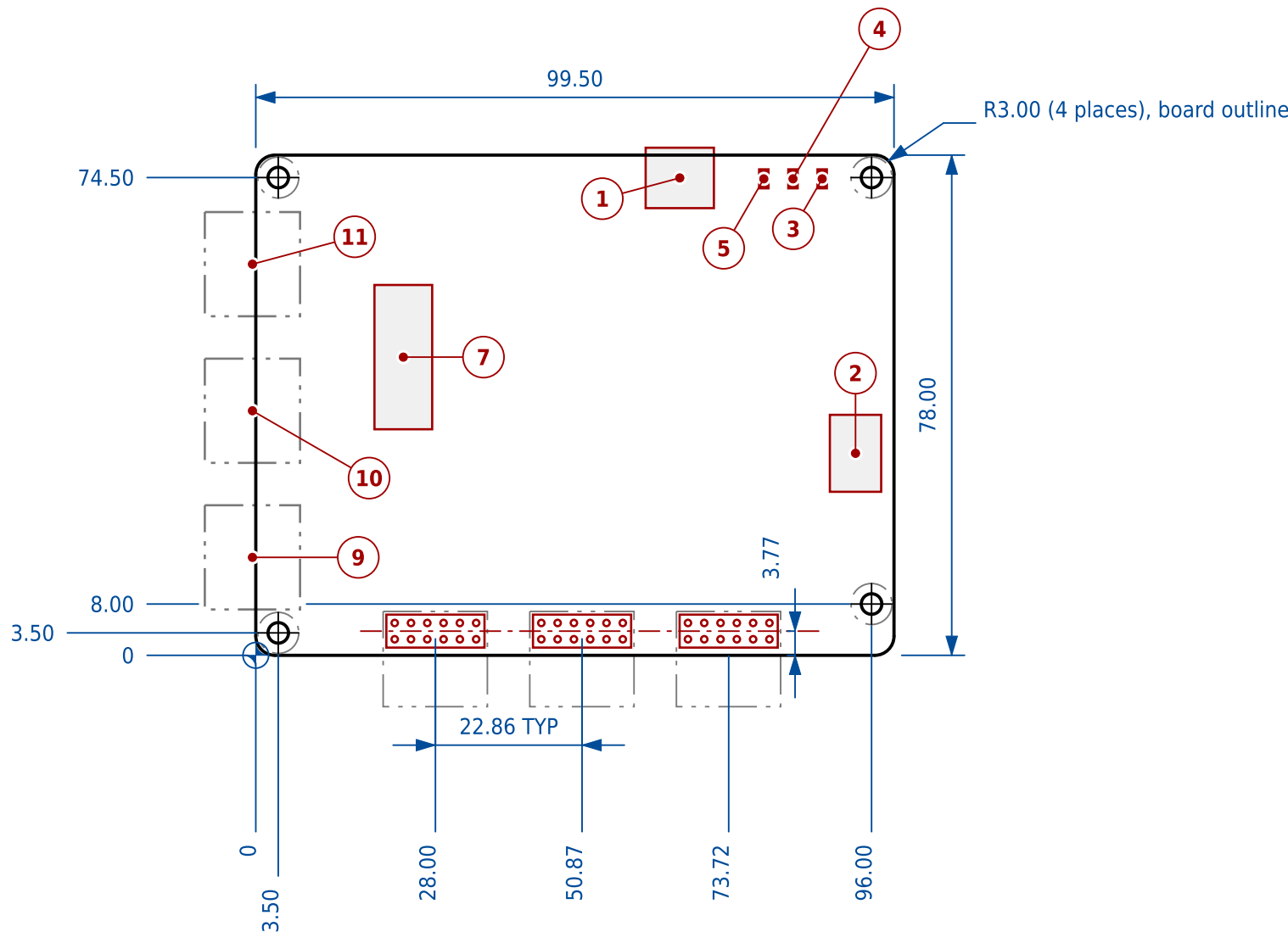
## LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Pmod connector body, hole keep-out
- Dimension, extension and leader

## DB 06+ V2.0.1 / DB 06+ V2.1.0

tinytapeout-demo rev 2.0.1 and 2.1.0

|                  |                   |                                                |       |
|------------------|-------------------|------------------------------------------------|-------|
| DRAWN            | VERSION           | UNITS                                          | SCALE |
| generated        | v0.0-98-g6a42ae3+ | mm                                             | 1:1   |
| SIZE             | SHEET             | DRAWING NO                                     | REV   |
| A3               | 1 OF 1            | TT-DB-03                                       | A     |
| MATERIAL         |                   | GENERAL TOLERANCE                              |       |
| PCB, 1.6 nominal |                   | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |       |



## NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
- Geometry is design nominal, read from the KiCad board file.
- Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
- No other demo board revision shares this geometry.
- The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0, is the same on every revision; their distance from the lower edge and their position along it are not.
- Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.
- Board thickness is 1.6 mm nominal. The KiCad file's 1.56252 mm is its stackup sum, not a specified finished thickness.
- Assembled envelope, connector overhang included, is 99.50 x 79.15 mm against a 99.50 x 78.00 mm board outline. The FEATURE SCHEDULE gives the extents.

## NOTES (continued)

- GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
- Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

## SOURCES

- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad\_pcb @ 028a51b1e - title block: TinyTapeout 06+ Demoboard rev 2.1.2, dated 2025-02-11
- Shuttle mapping: [https://docs.google.com/spreadsheets/d/1xD\\_uemuQcpUnXRY4twOrFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370](https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370) - row "DB 06+ v2.1.2", Used by: TT08.

## HOLE SCHEDULE

| ID  | X mm  | Y mm  | DIA mm | KEEPOUT mm |
|-----|-------|-------|--------|------------|
| MT1 | 3.50  | 3.50  | 3.20   | 6.40       |
| MT2 | 96.00 | 8.00  | 3.20   | 6.40       |
| MT3 | 3.50  | 74.50 | 3.20   | 6.40       |
| MT4 | 96.00 | 74.50 | 3.20   | 6.40       |

## PMOD HOST HEADERS

| PORT   | REF | EDGE   | CX mm  | CY mm | PIN1 X mm | PIN1 Y mm |
|--------|-----|--------|--------|-------|-----------|-----------|
| INPUT  | J3  | bottom | 28.005 | 3.775 | 34.355    | 5.045     |
| BIDIR  | J5  | bottom | 50.865 | 3.775 | 57.215    | 5.045     |
| OUTPUT | J6  | bottom | 73.725 | 3.775 | 80.075    | 5.045     |

## FEATURE SCHEDULE

| #  | FEATURE                                  | X EXTENT mm    | Y EXTENT mm    |
|----|------------------------------------------|----------------|----------------|
| 1  | USB-C power / control                    | 60.78 to 71.42 | 69.73 to 79.15 |
| 2  | 7-segment display                        | 89.50 to 97.50 | 25.50 to 37.50 |
| 3  | LED D1                                   | 87.57 to 89.03 | 72.82 to 75.78 |
| 4  | LED D2                                   | 83.02 to 84.48 | 72.82 to 75.78 |
| 5  | LED D3                                   | 78.47 to 79.93 | 72.82 to 75.78 |
| 6  | Fourth indicator LED - not on this board | -              | -              |
| 7  | 8-way input DIP switch                   | 18.50 to 27.50 | 35.25 to 57.75 |
| 8  | Second DIP switch - not on this board    | -              | -              |
| 9  | Side Pmod J12 (not fitted)               | -7.95 to 6.90  | 7.15 to 23.40  |
| 10 | Side Pmod J13 (not fitted)               | -7.95 to 6.90  | 30.01 to 46.26 |
| 11 | Side Pmod J14 (not fitted)               | -7.95 to 6.90  | 52.87 to 69.12 |

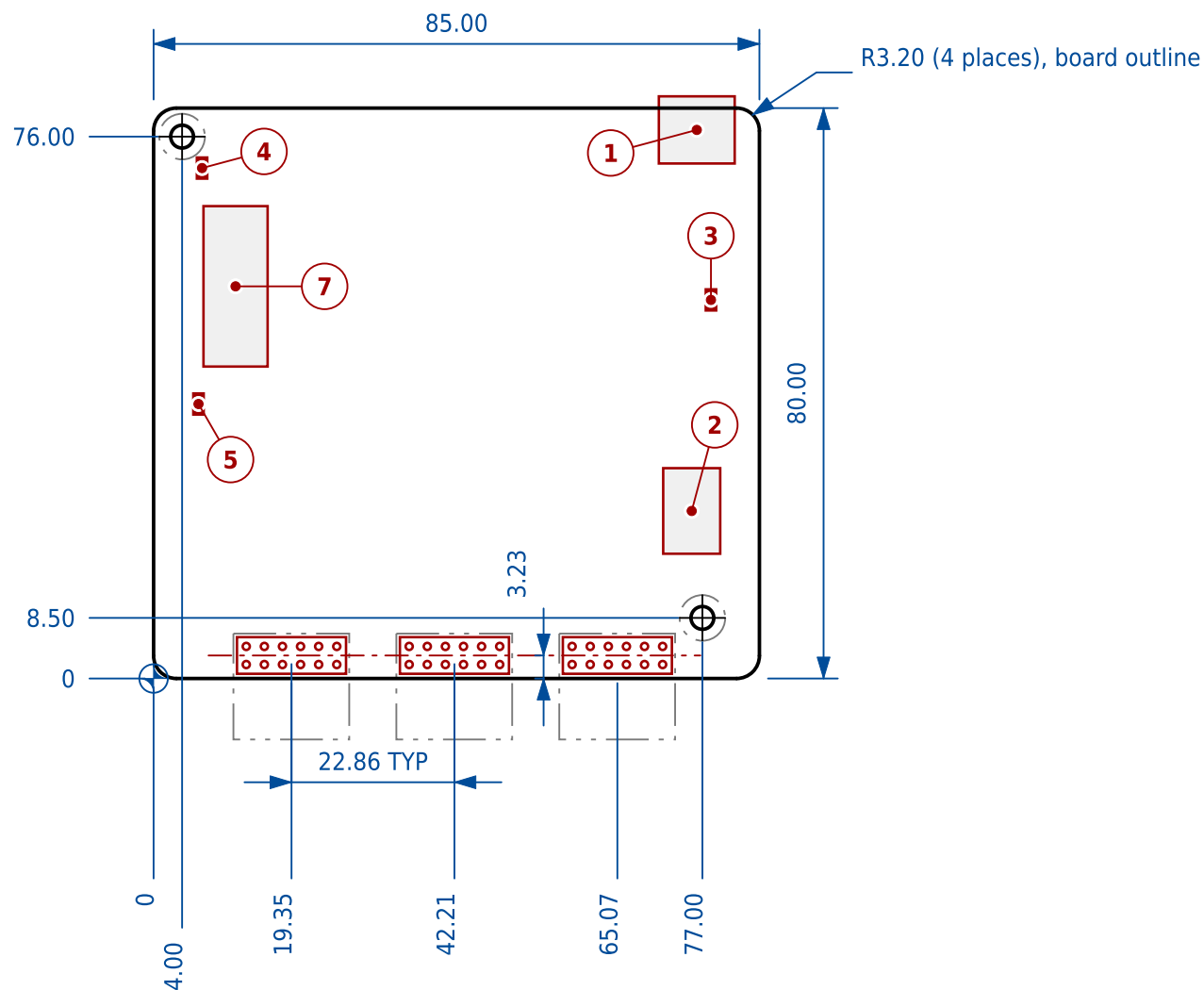
## LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Pmod connector body, hole keep-out
- Dimension, extension and leader

## DB 06+ V2.1.2

tinytapeout-demo rev 2.1.2

|                  |                   |                                                |       |
|------------------|-------------------|------------------------------------------------|-------|
| DRAWN            | VERSION           | UNITS                                          | SCALE |
| generated        | v0.0-98-g6a42ae3+ | mm                                             | 1:1   |
| SIZE             | SHEET             | DRAWING NO                                     | REV   |
| A3               | 1 OF 1            | TT-DB-04                                       | A     |
| MATERIAL         |                   | GENERAL TOLERANCE                              |       |
| PCB, 1.6 nominal |                   | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |       |



## NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
- Geometry is design nominal, read from the KiCad board file.
- Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
- No other demo board revision shares this geometry.
- ETR is Tiny Tapeout's own designation for this board. The KiCad title block reads "Tiny Tapeout Demoboard v3" and does not carry it; the name here is the board's ID in Tiny Tapeout's board revision spreadsheet. Nothing mechanical depends on it.
- The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0, is the same on every revision; their distance from the lower edge and their position along it are not.
- Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.

## NOTES (continued)

- Board thickness is 1.6 mm nominal. The KiCad file's 1.56100 mm is its stackup sum, not a specified finished thickness.
- Assembled envelope, connector overhang included, is 85.00 x 81.65 mm against a 85.00 x 80.00 mm board outline. The FEATURE SCHEDULE gives the extents.
- GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
- Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

## SOURCES

- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad\_pcb @ d830790ca - title block: Tiny Tapeout Demoboard v3 rev 3.2, dated 2025-11-12
- Shuttle mapping: [https://docs.google.com/spreadsheets/d/1xD\\_uemuQcpUnXRY4twOrFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370](https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370) - row "DB ETR v3.2", Used by: TT09, TTSKY25a, TTSKY25b, TTGF0p2.

## HOLE SCHEDULE

| ID  | X mm  | Y mm  | DIA mm | KEEPOUT mm |
|-----|-------|-------|--------|------------|
| MT1 | 4.00  | 76.00 | 3.20   | 6.40       |
| MT2 | 77.00 | 8.50  | 3.20   | 6.40       |

## PMOD HOST HEADERS

| PORT   | REF | EDGE   | CX mm  | CY mm | PIN1 X mm | PIN1 Y mm |
|--------|-----|--------|--------|-------|-----------|-----------|
| INPUT  | J11 | bottom | 19.350 | 3.230 | 25.700    | 4.500     |
| BIDIR  | J12 | bottom | 42.210 | 3.230 | 48.560    | 4.500     |
| OUTPUT | J13 | bottom | 65.070 | 3.230 | 71.420    | 4.500     |

## FEATURE SCHEDULE

| #  | FEATURE                                        | X EXTENT mm    | Y EXTENT mm    |
|----|------------------------------------------------|----------------|----------------|
| 1  | USB-C power / control                          | 70.89 to 81.53 | 72.23 to 81.65 |
| 2  | 7-segment display                              | 71.50 to 79.50 | 17.50 to 29.50 |
| 3  | LED D3                                         | 77.47 to 78.93 | 51.62 to 54.58 |
| 4  | LED D4                                         | 6.07 to 7.53   | 70.11 to 73.07 |
| 5  | LED D5                                         | 5.57 to 7.03   | 37.02 to 39.98 |
| 6  | Fourth indicator LED - not on this board       | -              | -              |
| 7  | 8-way input DIP switch                         | 7.00 to 16.00  | 43.75 to 66.25 |
| 8  | Second DIP switch - not on this board          | -              | -              |
| 9  | Side Pmod, first position - not on this board  | -              | -              |
| 10 | Side Pmod, second position - not on this board | -              | -              |
| 11 | Side Pmod, third position - not on this board  | -              | -              |

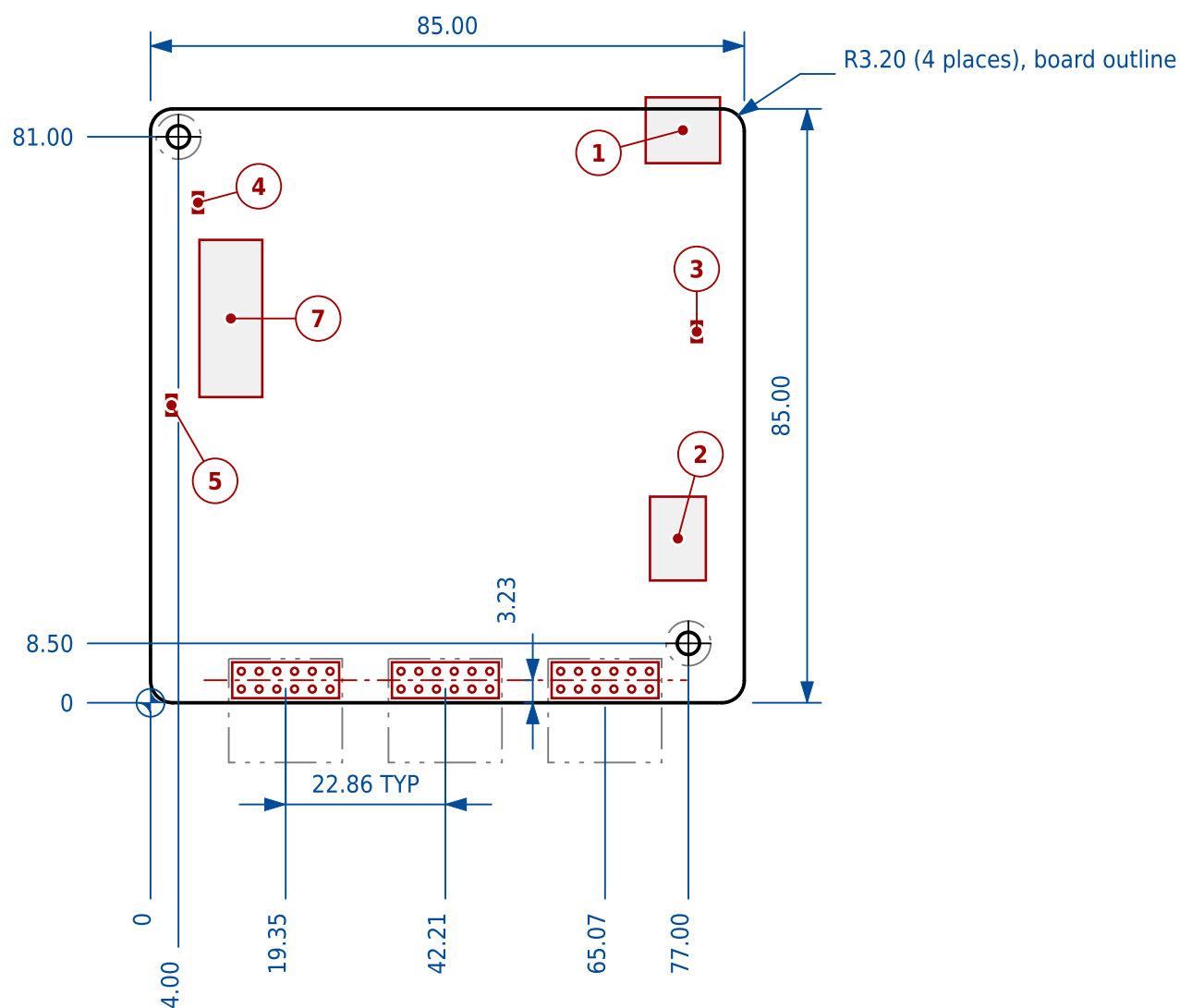
## LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body, hole keep-out
- Dimension, extension and leader

## DB ETR V3.2

tinytapeout-demo rev 3.2

|                  |                   |                                                |       |
|------------------|-------------------|------------------------------------------------|-------|
| DRAWN            | VERSION           | UNITS                                          | SCALE |
| generated        | v0.0-98-g6a42ae3+ | mm                                             | 1:1   |
| SIZE             | SHEET             | DRAWING NO                                     | REV   |
| A3               | 1 OF 1            | TT-DB-05                                       | A     |
| MATERIAL         |                   | GENERAL TOLERANCE                              |       |
| PCB, 1.6 nominal |                   | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |       |



## NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
- Geometry is design nominal, read from the KiCad board file.
- Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
- No other demo board revision shares this geometry.
- This revision has no row in Tiny Tapeout's board revision spreadsheet, which lists DB ETR v3.2 as the current demoboard. The name used here follows that ID; no shuttle is known to have shipped on it.
- The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0, is the same on every revision; their distance from the lower edge and their position along it are not.
- Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.
- Board thickness is 1.6 mm nominal. The KiCad file's 1.56100 mm is its stackup sum, not a specified finished thickness.

## NOTES (continued)

- Assembled envelope, connector overhang included, is 85.00 x 86.65 mm against a 85.00 x 85.00 mm board outline. The FEATURE SCHEDULE gives the extents.
- GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
- Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

## SOURCES

- KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad\_pcb @ ecb636ace - title block: Tiny Tapeout Demoboard v3 rev 3.3, dated 2026-06-23
- Shuttle mapping: [https://docs.google.com/spreadsheets/d/1xD\\_uemuQcpUnXRY4twOrFzF0urJLVWMdqAnrOJLetM/edit?gid=1250963370](https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrFzF0urJLVWMdqAnrOJLetM/edit?gid=1250963370) - no row: this revision is not in the board revision spreadsheet, so no shuttle is recorded for it.

## HOLE SCHEDULE

| ID  | X mm  | Y mm  | DIA mm | KEEPOUT mm |
|-----|-------|-------|--------|------------|
| MT1 | 4.00  | 81.00 | 3.20   | 6.40       |
| MT2 | 77.00 | 8.50  | 3.20   | 6.40       |

## PMOD HOST HEADERS

| PORT   | REF | EDGE   | CX mm  | CY mm | PIN1 X mm | PIN1 Y mm |
|--------|-----|--------|--------|-------|-----------|-----------|
| INPUT  | J11 | bottom | 19.350 | 3.230 | 25.700    | 4.500     |
| BIDIR  | J12 | bottom | 42.210 | 3.230 | 48.560    | 4.500     |
| OUTPUT | J13 | bottom | 65.070 | 3.230 | 71.420    | 4.500     |

## FEATURE SCHEDULE

| #  | FEATURE                                        | X EXTENT mm    | Y EXTENT mm    |
|----|------------------------------------------------|----------------|----------------|
| 1  | USB-C power / control                          | 70.89 to 81.53 | 77.23 to 86.65 |
| 2  | 7-segment display                              | 71.50 to 79.50 | 17.50 to 29.50 |
| 3  | LED D3                                         | 77.47 to 78.93 | 51.62 to 54.58 |
| 4  | LED D4                                         | 6.07 to 7.53   | 70.11 to 73.07 |
| 5  | LED D5                                         | 2.27 to 3.73   | 41.11 to 44.07 |
| 6  | Fourth indicator LED - not on this board       | -              | -              |
| 7  | 8-way input DIP switch                         | 7.00 to 16.00  | 43.75 to 66.25 |
| 8  | Second DIP switch - not on this board          | -              | -              |
| 9  | Side Pmod, first position - not on this board  | -              | -              |
| 10 | Side Pmod, second position - not on this board | -              | -              |
| 11 | Side Pmod, third position - not on this board  | -              | -              |

## LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body, hole keep-out
- Dimension, extension and leader

## DB ETR V3.3

tinytapeout-demo rev 3.3

|                  |                   |                                                |       |
|------------------|-------------------|------------------------------------------------|-------|
| DRAWN            | VERSION           | UNITS                                          | SCALE |
| generated        | v0.0-98-g6a42ae3+ | mm                                             | 1:1   |
| SIZE             | SHEET             | DRAWING NO                                     | REV   |
| A3               | 1 OF 1            | TT-DB-06                                       | A     |
| MATERIAL         |                   | GENERAL TOLERANCE                              |       |
| PCB, 1.6 nominal |                   | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |       |